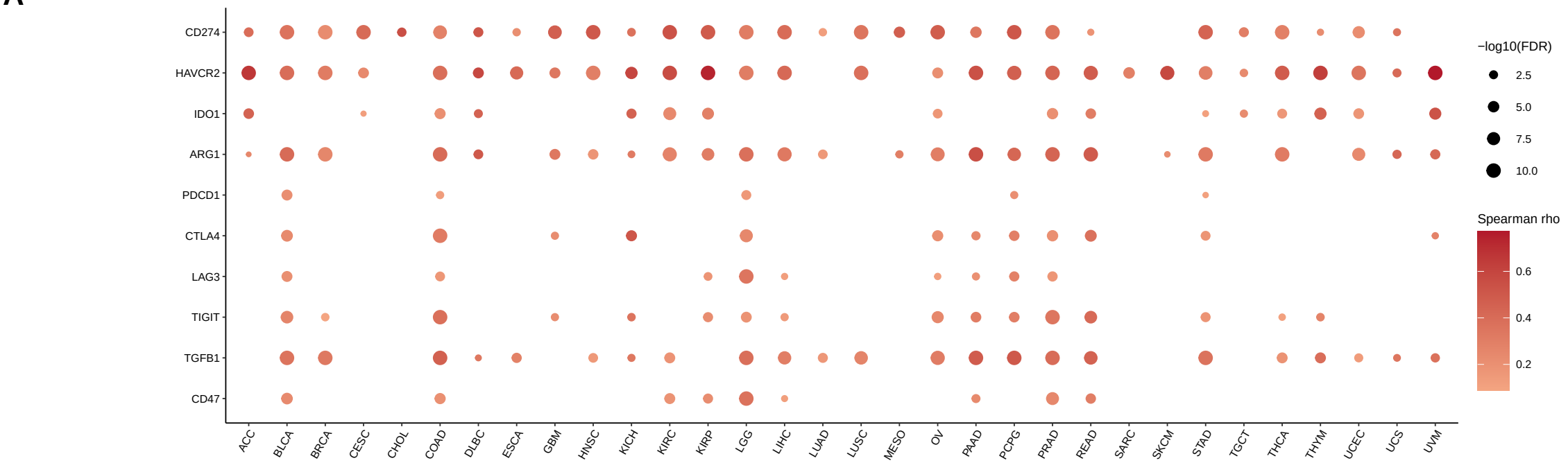
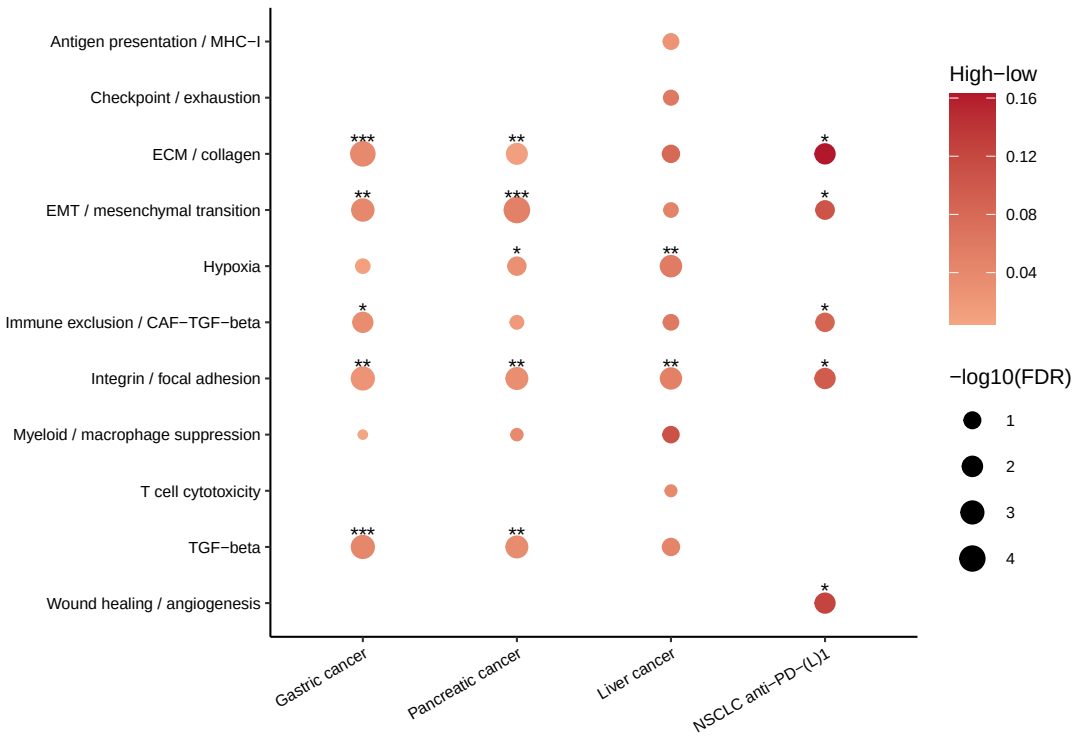
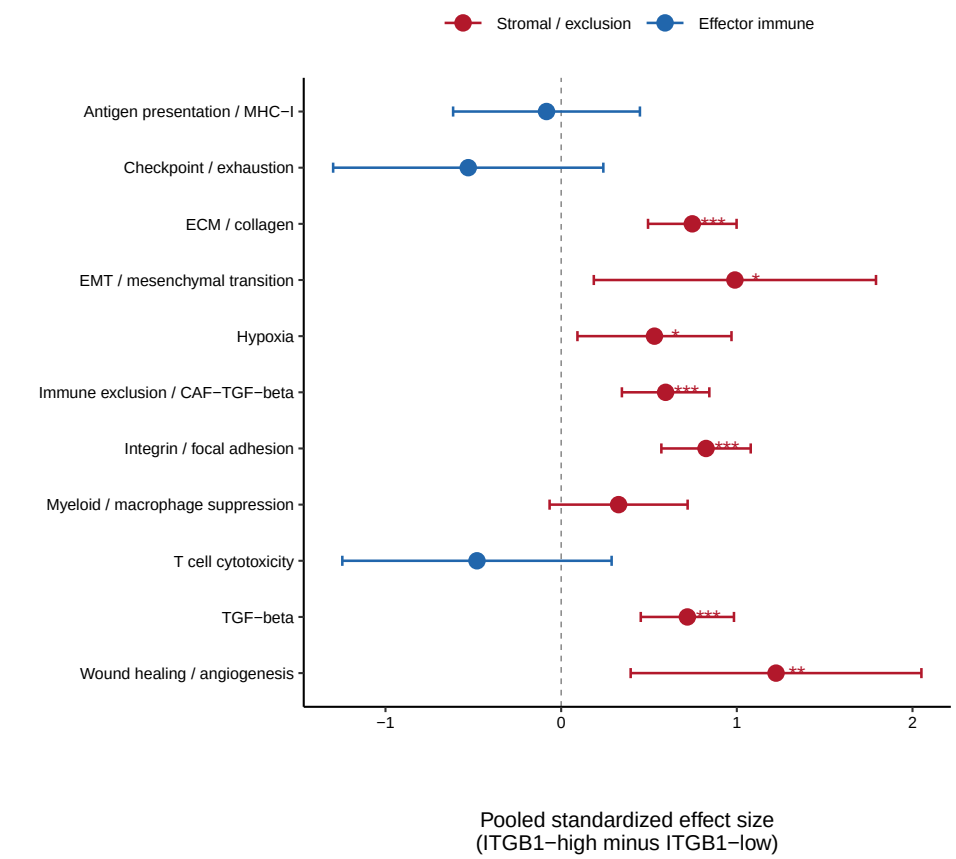
A


CD274
HAVCR2
IDO1
ARG1
PDCD1
CTLA4
LAG3
TIGIT
TGFB1
CD47
ACC
BLCA
BRCA
CESC
CHOL
COAD
DLBC
ESCA
GBM
HNSC
KICH
KIRC
KIRP
LGG
LIHC
LUAD
LUSC
MESO
OV
PAAD
PCPG
PRAD
READ
SARC
SKCM
STAD
TGCT
THCA
THYM
UCEC
UCS
UVM
−log10(FDR)
2.5
5.0
7.5
10.0
Spearman rho
0.6
0.4
0.2

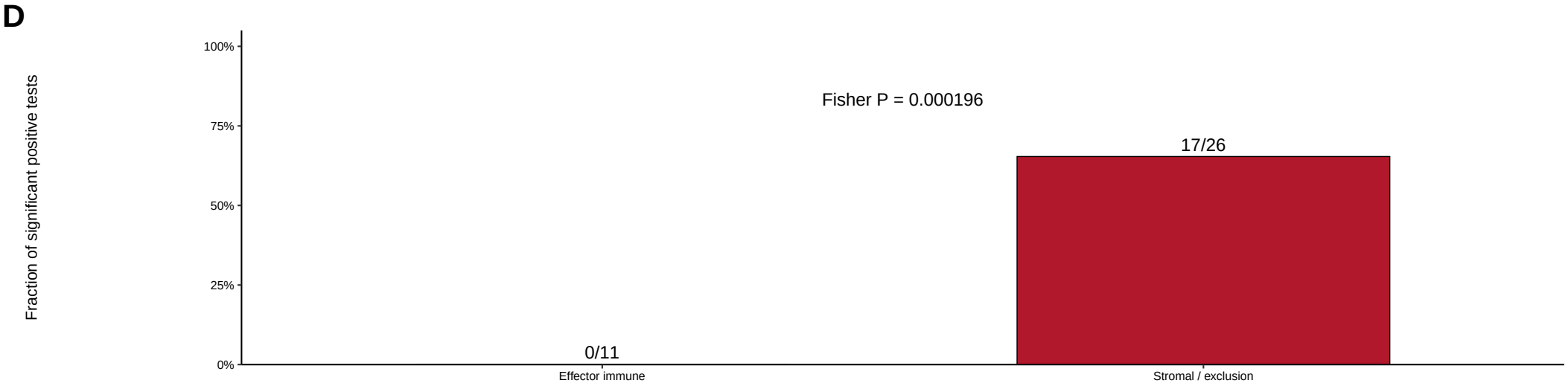
B


Antigen presentation / MHC−I
Checkpoint / exhaustion
ECM / collagen
EMT / mesenchymal transition
Hypoxia
Immune exclusion / CAF−TGF−beta
Integrin / focal adhesion
Myeloid / macrophage suppression
T cell cytotoxicity
TGF−beta
Wound healing / angiogenesis
Gastric cancer
Pancreatic cancer
Liver cancer
NSCLC anti−PD−(L)1
High−low
0.16
0.12
0.08
0.04
−log10(FDR)
1
2
3
4

C


Stromal / exclusion
Effector immune
Antigen presentation / MHC−I
Checkpoint / exhaustion
ECM / collagen
EMT / mesenchymal transition
Hypoxia
Immune exclusion / CAF−TGF−beta
Integrin / focal adhesion
Myeloid / macrophage suppression
T cell cytotoxicity
TGF−beta
Wound healing / angiogenesis
−1
0
1
2
Pooled standardized effect size
(ITGB1−high minus ITGB1−low)

D


Fraction of significant positive tests
100%
75%
50%
25%
0%
Fisher P = 0.000196
17/26
0/11
Effector immune
Stromal / exclusion